Wireshark, DNS and HTTP/HTTPS Assignment

1. Wireshark Traffic Capture

Website: www.spotify.com
DNS Packet 1: _____
DNS Packet 2: _____
HTTPS Packet 1: _____
HTTPS Packet 2: _____
Attach Wireshark screenshots and .pcap file.

2. nslookup for www.flipkart.com

Command: nslookup www.flipkart.com
Returned IP Address: _____
The result shows the IP address that the domain name resolves to and helps verify DNS functionality.

3. HTTP vs HTTPS

1. HTTP is unencrypted while HTTPS uses SSL/TLS encryption.
2. HTTP uses port 80 while HTTPS uses port 443.
3. HTTPS protects data from interception and tampering.

Example: Instagram, Paytm, YouTube and Flipkart use HTTPS.

4. DNS Query and Response Analysis

Filter: dns
Typical sequence:
1. Standard Query for a domain.
2. DNS server receives request.
3. Standard Query Response returns IP address.
4. Browser connects to returned IP.

Your observations: _____

5. Protocol Scenario (DHCP Example)

When a phone connects to cafe WiFi before opening Instagram, DHCP automatically assigns an IP address, subnet mask, gateway and DNS settings so the device can communicate on the network.

Submission Notes

Attach Wireshark screenshots, nslookup output screenshot, and your packet observations from the actual lab activity.